

ee Hello world!"

Introduction to python

→ Hey, But why python? (High level)

→ All product companies have requirement for python expertise (OA+DS)

✓ → AI is powered by Python

✓ → DAV, ML, NLP, DL, CV all depends on this language

→ Easy Simple language (IP)

→ works on all datatypes

✓ [Text, img, video, Audio] etc

Total session (Python 1) → 11

→ 1. Data types, variables & Input/output

2. operators

3. Control statement (if-else)

4 & 5. Loops in python (2 sessions)
→ Automation

6. functions

= sum ()

738 → list in python (Data structure)
↑↑

9 → strings

10 → sets and tuple

11 → dictionary

→ "Module test will be there"

"Don't skip a single session!"

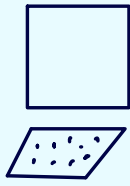
Session 1

Agenda

1. why programming language?
2. Getting started with Python
3. print statement
4. Data Types
 - Integer
 - Boolean
 - float
 - String
 - None
5. Type function
6. Variable Names
7. Input/output

Hi, Heyyy!

How do we communicate with Computer?



→ what language this Machine understand?



[0,1] → Binary

India



[Modiji]



[Hindi]

Italy

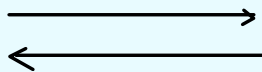


Meloniji



Italian

Translator



Translator / Interpreter



(Human)

(computer)

[Program (text file)]



→
Rules

Python

Interpreter
app



Box

Running
program



syntax

- 1 A **program** is written in a programming language.
- 2 In order for the program to work, code must follow the **syntax** rules of the language.
- 3 A special app — an **interpreter** — understands code and actually launches the program (makes it alive).

[cricket watch I like to] input
↓

I like to watch cricket

→ Python is open source
↓

→ 3 way → CMD → write
→ compile
→ save.py ↓
- Run

IDE → Integrated Dev Env

↓
online offline — VS Code
↓ ↓
 ↓
 ↓
 ↓
 pycharm

Text string $\rightarrow$ [ee 00
or
c 2]

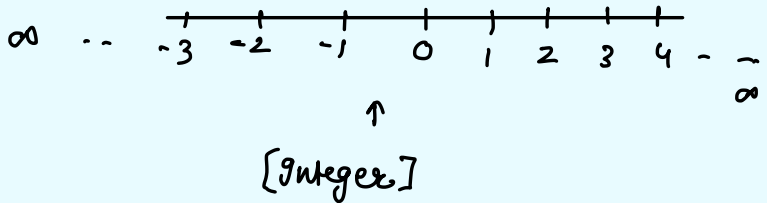
Later $\rightarrow$ eee 000

Datatype

5 Main datatypes

1. String $\rightarrow$ ee 00 or e 2

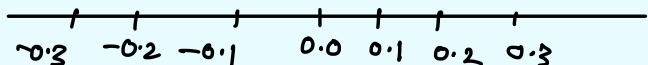
2. Integer $\rightarrow$ positive or neg whole number



3. Decimal $\rightarrow$ float

.9 $\rightarrow$ float

Real
number $\rightarrow$



9.0	vs	9	→ No are same
↓		↓	
float		int	

4. Boolean → True | False

9 > 5	} → <u>Boolean</u>
→ True	
2 > 3	
→ false	

cc Object oriented PL ""

↓

Name → Table → object

↓

[space, location]

print("Hello")

↓

variable

↙

name

ph-no

↗

9112

assign

→ object

$$\rightarrow \underline{1144251698} + \underline{1144251698}$$

$$\begin{array}{c} x = \underline{1144251698} \\ \uparrow \quad \uparrow \quad \uparrow \\ \text{VN} \quad \text{Assign} \quad \text{object} \end{array}$$

$$\rightarrow \underline{x+x}, \quad \underline{x \times 8}$$

Rules while giving variable Name

1 $\rightarrow$ A-Z, 0-9, _ $\rightarrow$ underscore

Hi-Hello
 $\rightarrow$ x

Hi#

2. It should never ever start with a Number

$$\rightarrow \underline{1a = 100} \quad \times \text{ error}$$

$$\begin{array}{c} a1 = 100 \quad \checkmark \\ \uparrow \end{array}$$

letter

$$-1 = 100 \quad \checkmark$$

3. we cannot use space in variable Name

$x_y = 100$
↳ underscore

4. $x = 'hello'$
 x = 'Hi' → Two diff variable
↓
Case sensitive

5 → Never assign any default variables/
functions as variable name

[print = 5] ✓
↑ ↑

print('Hello')

UI → Anaconda

